GUNJAN PRAMOD RAJGURE

AI/ML Engineer | LLM Systems & RAG | MLOps & Evaluation | Biomedical AI

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Portfolio Website | Github | Linkedin

SUMMARY

AI/ML engineer with 4+ years building **LLM infrastructure**, **retrieval-augmented generation**, **MLOps**, and **biomedical NLP** in **Python**, each measured against a baseline, combining research rigor with enterprise delivery of production APIs and evaluation pipelines.

EDUCATION & CERTIFICATIONS

M.S. Bioinformatics and Data Science, University of Delaware | GPA: 3.86/4.00 | May 2026

B.Tech. Computer Science and Engineering, DKTE Textile and Engineering Institute | June 2021

Google Advanced Data Analytics Specialization, Coursera, 2024

PROFESSIONAL EXPERIENCE

Research Assistant, Applied AI Engineering | University of Delaware May 2025 - Present

- Architected **PhenoZebra**, a multimodal rare-disease **knowledge graph** fusing **PyTorch** text and image **embeddings** across 69,003 clinical cases and surfacing 711,264 candidate relationships.
- Fine-tuned **BERT** relation extraction for **eMIND**, reaching ~0.94 precision and ~0.84 recall on expert-annotated mutation-impact relations, served in a live **evidence search** portal.
- Optimized **NVIDIA A100** inference with bfloat16, batching, caching, and checkpoint recovery, and normalized 50,000+ PubMed Central articles through **Python** pipelines with **data quality** checks.

Systems Engineer, ML & Enterprise Systems | Tata Consultancy Services July 2021 - July 2024

- Led an eight-member team delivering an **NLP** classification system over 100,000+ records at 95% accuracy with **TensorFlow** and **scikit-learn**, from requirements to production.
- Delivered 50+ **REST APIs** and enterprise applications secured with **Keycloak**, deployed with **Docker** and **Kubernetes** and supported in production with **observability** dashboards.
- Automated **model evaluation**, scoring, and reporting workflows in **Python** and **MLflow**, cutting manual effort 60% and catching regressions before release.
- Developed **ranking** models evaluated by **NDCG** with **Elasticsearch** search, shipped through **Jenkins** CI/CD with **pytest** and SonarQube quality gates.
- Translated client requirements into technical designs and solution architecture, documented releases, and supported **A/B testing** and **forecasting** analyses for pricing.

PROJECTS

LLM Inference Gateway | github.com/GunjanPR/01-llm-gateway

- Engineered an OpenAI-compatible **LLM gateway** (**FastAPI**, **Redis**, **Prometheus**, **OpenTelemetry**) with routing, semantic caching, and PII guardrails; load shedding raised goodput at 3x overload from 1.9 to 19.3 req/s on a mock backend.

Citation-Grounded Medical RAG | github.com/GunjanPR/02-healthcare-rag

- Created hybrid **BM25** and dense retrieval with **MedCPT** reranking in **LangGraph** over 500 PMC articles, lifting NDCG@10 24% over BM25 on 92 held-out questions, with 95.3% of answers faithful by blind hand labels.

SKILLS

- **LLM Systems & RAG:** LLM gateways, RAG, LangGraph, hybrid retrieval, OpenSearch, Qdrant, reranking, semantic caching, guardrails, Ollama, vLLM, BERT, Transformers, LLM evaluation
- **Machine Learning & Statistics:** PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Optuna, SHAP, time series forecasting, bootstrap confidence intervals, significance testing, calibration
- **MLOps & Infrastructure:** MLflow, DVC, Evidently, Docker, Kubernetes, GitHub Actions, CI/CD, FastAPI, Redis, PostgreSQL, Prometheus, Grafana, OpenTelemetry, AWS, Azure, GCP
- **Biomedical AI & Data:** biomedical NLP, entity normalization, ontologies (HPO, DOID), FHIR, multimodal embeddings, NVIDIA A100, Python, SQL, Java, REST APIs, Keycloak